

Sentiment Analysis Report

Project Name: Amazon Customer Review Sentiment Analysis

Internship: CodeAlpha Data Analytics Internship

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1. Business Problem

With the rapid growth of e-commerce, customers share thousands of reviews every day after purchasing products. These reviews contain valuable information about customer satisfaction, product quality, and common issues. However, manually reading and analyzing such a large volume of reviews is time-consuming and impractical.

As a result, businesses often struggle to quickly identify customer concerns, understand overall sentiment, and make timely product improvements. An automated sentiment analysis system can help businesses classify customer reviews into Positive, Negative, and Neutral sentiments, allowing them to monitor customer feedback efficiently and make data-driven business decisions.

2. Business Objectives

- Classify reviews into Positive, Negative and Neutral sentiments.
- Measure customer satisfaction.
- Identify frequently discussed product features.
- Detect recurring customer complaints.
- Evaluate a machine learning model.
- Generate business insights and recommendations.

3. Dataset Overview

This project uses the **Amazon Electronics Reviews** dataset obtained from Kaggle. The dataset contains customer reviews submitted for various electronic products sold on Amazon. Each review includes important information such as the review text, review summary, customer rating, product ID, reviewer details, and review date.

Since machine learning algorithms cannot directly understand textual data, the review text was cleaned and transformed into numerical features using Natural Language Processing techniques before model training.

4. Data Preprocessing

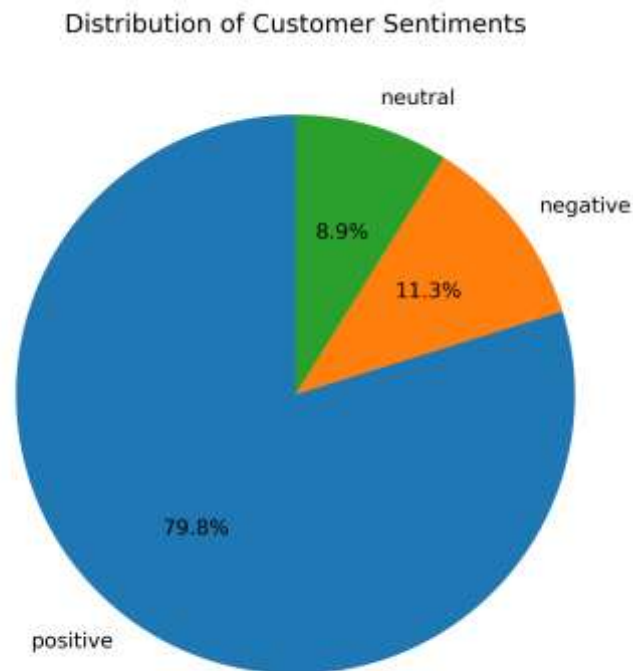
- **Renamed column names for better readability.**
- **Converted all review text to lowercase.**
- **Removed punctuation marks and special characters.**
- **Removed stopwords such as *the*, *is*, and *and*.**
- **Tokenized the review text into individual words.**
- **Converted cleaned text into numerical features using Bag of Words and TF-IDF Vectorization.**
- **Generated sentiment labels from customer ratings.**

5. Sentiment Analysis

5.1 Sentiment Distribution Analysis

Observation: The sentiment distribution shows that approximately **79.8%** of customer reviews are Positive, while only a small percentage are Negative or Neutral. This indicates that most customers were satisfied with the products they purchased.

Visualization:



Key Finding:

- Positive reviews dominate the dataset.
- Negative reviews account for a relatively small proportion of customer feedback.
- Neutral reviews are the least common.

Business Insight: The overall customer satisfaction level appears to be high. However, businesses should not ignore the smaller percentage of negative reviews, as they often highlight product issues that directly impact customer experience

5.2 Model Performance Analysis

Visualization:

Accuracy : 0.824

Observation: The Logistic Regression model achieved an overall accuracy of approximately **82.4%**, indicating that it successfully classified most customer reviews into the correct sentiment category.

Key findings : The model demonstrated good overall performance despite the dataset being imbalanced. It performed particularly well in identifying Positive reviews.

Business Insight : This result demonstrates that sentiment analysis can significantly reduce the manual effort required to analyze thousands of customer reviews while providing reliable insights into customer satisfaction.

5.3 Confusion Matrix Analysis

Visualization:

```
[[ 56   1 173]
 [  6   4 162]
 [  7   3 1588]]
```

Observation: The confusion matrix shows that the model correctly classified most Positive reviews. However, many Neutral reviews were predicted as Positive, indicating that the model had difficulty distinguishing between these two sentiment categories.

Key findings:

- Positive sentiment achieved the highest prediction accuracy.
- Neutral reviews were the most challenging to classify.
- Class imbalance affected model performance.

Business Insight: Future improvements should focus on collecting more Neutral reviews or applying class-balancing techniques to improve prediction performance across all sentiment categories.

5.4 Classification Report Analysis

Visualization:

	precision	recall	f1-score	support
negative	0.81	0.24	0.37	230
neutral	0.50	0.02	0.04	172
positive	0.83	0.99	0.90	1598
accuracy			0.82	2000
macro avg	0.71	0.42	0.44	2000
weighted avg	0.80	0.82	0.77	2000

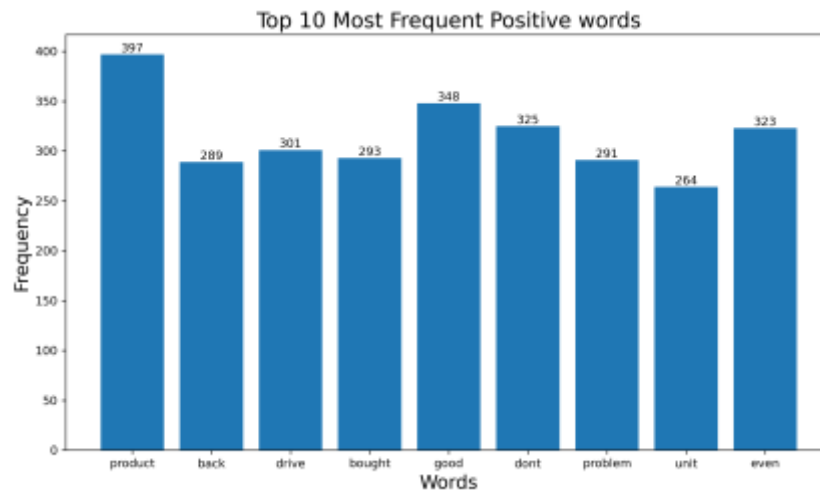
Observation: The classification report indicates that the model achieved excellent Precision, Recall, and F1-score for Positive reviews. However, the performance for Neutral reviews was comparatively lower.

Key findings : Although the overall accuracy is high, Precision, Recall, and F1-score provide a more detailed understanding of model performance. These metrics reveal that the model is more reliable for predicting Positive reviews than Neutral reviews

Business Insight: Accuracy should be interpreted alongside Precision, Recall and F1-score.

5.5 Positive Review Analysis

Visualization:



Observation: The positive review analysis shows that words such as **product, good, don't, even, drive, bought, problem, unit, and back** appear frequently in positive customer reviews. This indicates that customers often discuss the product itself, their purchasing experience, and specific product features while expressing positive opinions.

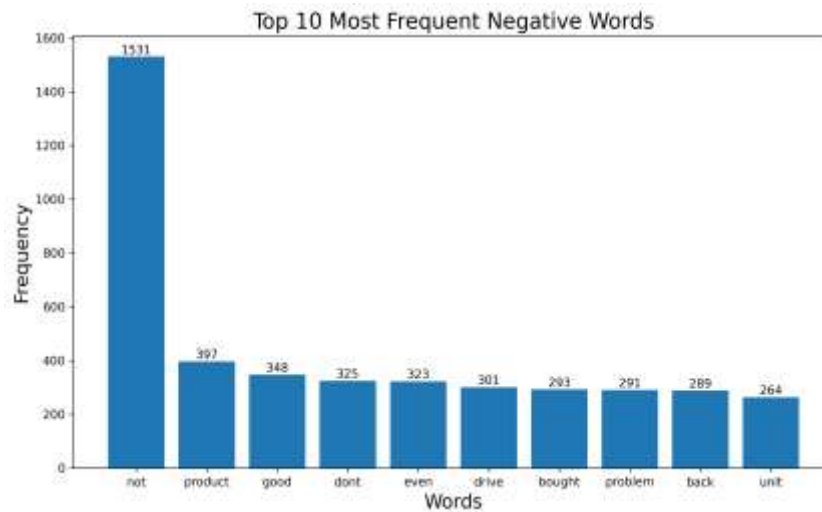
Key findings :

- The word **product** is the most frequently occurring term in positive reviews.
- Words such as **good** indicate that customers generally have a favorable opinion about the product.
- Other frequently appearing words, including **drive, bought, unit, and back**, suggest that customers often describe their purchase experience and product usage in their reviews.
- Some common conversational words such as **don't** and **even** are still present because they occur frequently in natural customer language.
-

Business Insight: Positive reviews indicate that customers are generally satisfied with the products they purchase. Businesses can further analyze the context of frequently occurring product-related words to understand which product features contribute most to customer satisfaction and use these strengths in future marketing strategies.

5.6 Negative Review Analysis

Visualization :



Observation: The negative review analysis shows that the word **not** appears far more frequently than any other word, followed by **product**, **good**, **don't**, **even**, **drive**, **bought**, **problem**, **back**, and **unit**. The high occurrence of the word **not** indicates that dissatisfied customers frequently use negative expressions while describing their experience.

Key findings:

- **Not** is the most dominant word in negative reviews, reflecting strong expressions of dissatisfaction.
- Words such as **problem**, **unit**, and **product** suggest that customers commonly discuss product-related issues.
- The presence of words like **drive**, **back**, and **bought** indicates that customers often describe product defects, returns, replacements, or purchase experiences in their complaints.
- The negative reviews focus more on problems experienced after purchasing the product than on specific product features.

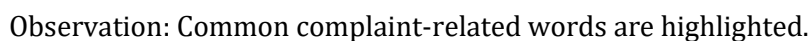
Business Insight: The company should investigate reviews containing words such as **problem**, **unit**, and **not** to identify the root causes of customer dissatisfaction. Addressing these recurring issues can improve product quality, reduce customer complaints, and enhance overall customer satisfaction.

Visualization:



5.8 Negative Word Cloud Analysis

Visualization:



Business Insight: Helps prioritize recurring customer issues

Overall Key Findings

- Approximately **79.8%** of customer reviews expressed Positive sentiment, indicating high customer satisfaction.
- The Logistic Regression model achieved an overall accuracy of approximately **82.4%**, demonstrating good sentiment classification performance.
- Product features such as **camera, battery, sound, quality**, and **price** were frequently appreciated by customers.
- Negative reviews commonly highlighted product-related issues through words such as **problem, unit**, and **cable**.
- Word frequency analysis and Word Clouds provided valuable insights into customer preferences and recurring concerns.

Business Recommendations

- Continue maintaining and promoting the product features that customers appreciate most, such as camera quality, battery performance, sound quality, and overall product quality.
- Investigate recurring customer complaints to identify the root causes of product issues.
- Strengthen product quality assurance and testing processes before shipment to reduce defective products.
- Regularly monitor customer sentiment using automated sentiment analysis to identify emerging issues at an early stage.
- Use positive customer feedback to support marketing campaigns while using negative feedback to guide product improvements.

Conclusion

This project successfully demonstrated the use of Natural Language Processing (NLP) and Machine Learning techniques to analyze customer opinions from Amazon product reviews. After performing text preprocessing and feature extraction using TF-IDF, a Logistic Regression model was developed to classify reviews into Positive, Negative, and Neutral sentiments.

The model achieved an overall accuracy of approximately **82.4%**, indicating that machine learning can effectively automate sentiment analysis for large volumes of customer feedback. Beyond prediction, the project also provided meaningful business insights by identifying frequently appreciated product features and recurring customer complaints.

Overall, this project highlights how sentiment analysis can help businesses better understand customer opinions, improve product quality, enhance customer satisfaction, and support data-driven decision-making.